

Foundations of Cyber Security

COMSM 0118

MSc in Cyber Security

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September October 2024

Unit Learning

Course Materials: Teams site.

Labs: The computers you are using. Everything is set up.

Assessment: Blackboard

Key topics to be covered

- ~~Objectives and needs of cyber security and fundamental principles~~
 - ~~Web Security (to serve as a case study)~~
 - ~~Risk analysis and threat modelling~~
 - Introduction to legal and ethical aspects
 - Applied cryptography
 - Fundamentals of security operations and incident management
- Covered in
Week 1

Learning Objectives

- A broad understanding of how cyber security interacts with the law and the ethical issues surrounding the discipline.
- A practical knowledge of cryptographic tools (e.g., asymmetric and symmetric encryption, password storage, digital signatures, FIPS-140 standards).
- Ability to apply knowledge to provide fundamental security in real-world applications.

Assessment

Coursework (100%)

(Group work: 40%) Threat modelling and risk analysis exercise on a case study

(Individual work: 60%) Implementing security principles and design practices to develop a secure application using off-the-shelf security and cryptography libraries and APIs

Assessment Deadline

Individual Assessment: 13:00 Friday 11 October 2024

Structure of Week 3

Monday 9 October 2023

Timeslot (approx).	Topic	Type
09:00-09:20	Welcome, feedback on the group work, Overview of the week, topics to be covered, activities, labs, Q&A	Lecture
09:20-09:50	Introduction to law and regulations	Lecture
10:00-10:50	Vulnerability research	Lecture
11:00-11:50	Guided discussion around security practitioners and regulation.	Roundtable Discussion
12:00-13:00	Lunch	
13:00-13:50	Ethics	Lecture
14:00-14:50	Privacy and Human Rights	Lecture
15:00-17:00	Group work: Students working on client-side scanning case.	Student presentation

Tuesday 10 October 2023

Cryptography

Wednesday 11 October 2023

Cryptography

Thursday 12 October 2023

Cryptography

Friday 13 October 2023

Fundamentals of security operations and incident management

Ethics and Responsible Disclosure

As future cyber security professionals, you are going to learn about how to break systems as part of this unit and following units

You must use this knowledge responsibly

At no point should you use it for unethical or illegal actions

The university network is not a lab, nor are any public servers or anyone else's machines/devices

You are provided with an extensive environment in this lab to try things out and learn about a range of attacks and defences

If at any point you identify a vulnerability, you must follow responsible disclosure practices. Please get in touch with the course lecturer immediately on how to do this.

Questions